

VSC80 / VSC800 Commands

FTDI - USB interface to PC. Open the FTDI driver by Serial Number "VSC80" or "VSC800".

Where possible, a 3 character V8 mnemonic is given in BLUE. Either code can be used. In general, new VSC80 commands have an intensity value following the instruction. If this is omitted, the previous intensity is used. V8 commands can use, for example Rs+ (fully on), Rs- (off), Rs=n (intensity=n). The intensity range n is 0 (off) to 1023 (fully on).

Intensity can also be set with a separate instruction, however there are eight(!) intensity channels (2 on the transmitted PCB, 1 on the lamp tray PCB, 1 on the spot PCB and 2 each on left and right flood PCBs)

In general, the V6/V8 did not have (this amount of) dimming on all channels, so these may be additional instructions. Intensities will be set to 50% (512) at power on.

The Commands listed below (case insensitive) should be terminated with a line feed (0x0a) character. Commands and argument(s) may be separated by a space, comma or '=' character.

On acceptance of the command, a newline character is returned. Some commands preceded this with the character '0' (success) or '1' (fail), or the requested data. A command which is not understood will return "What is " followed by the command.

Internal PC

If the internal PC is installed and connected, the VSC80 will bleep twice on pressing the front panel power switch to ON, otherwise it will bleep once, indicating it is to be controlled by an external PC. The firmware will then start. If neither PC is detected, there will be three bleeps, you will need to cycle the VSC80 power after reconnecting.

If the power switch is held down, there will be a series of rapid warning bleeps, at the end of this there will be a long bleep, the VSC80 will then immediately power down.

If the internal PC is shut down from Windows, the VSC80 will power down, after a long bleep, usually 10-20 seconds after the monitor signal is lost.

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VSC80 Command list

General commands

OFF (lf#)

Turns all LEDS/Lights off.

BUILD (Vs?)

Returns the VSC80 controller firmware build number.

TYPE

Returns machine type (VCS80.1, VSC800.1, etc)

SLAVE (Vv?)

Returns the slave controller (lamp tray) firmware build number.

DATE (Vd?)

Returns the VSC80 controller firmware build date.

ENUM (En?)

Returns a comma separated list of connected PCBs by interrogating the I2C bus: LAMPTRAY, LFLOOD, RFLOOD, SPOT, TRANS, ASTOKES, ZOOM, EEPROM, KEYPAD, ERROR, TEMP (temperature sensor on the spot PCB)

Note devices are only available for use after a main controller reset. (which initialises the I/O devices on the PCBs). If ERROR appears in the list, there is a possible I2C bus error.

DV?

Returns 170 if the trans base has not been removed since power on, else returns something else. If the trans base is removed and then replaced, the returned value is reset to 170 after the next successful trans light on command.

DV

If DV? Does not return 170, send this command to re-initialise the tray. If successful it returns 0, Dv? Should then return 170, else the tray has not been re-fitted.

I2C 70

Will return 0 if the trans base is connected or non-zero if disconnected. This writes to the tray I2C address (70) and returns the Acknowledge response.

SSTATUS (S??) (motor fault bits set if failed to find the home sensor)

Returns the status of the motors.

0x01 Camera filter motors not yet home.

0x02 V 80 Close up lens motor not yet home. (Back Focus in V800)

0x04 One or more motors are moving.

0x08 V 80 Coax motor not yet home. (Not used in V800)

0x10 Camera filter motor fault.

0x20 V 80 Close up lens motor fault. (Back Focus in V800)

0x80 V 80 Coax motor fault. (Not used in V800)

LSTATUS (Y??)

Returns the flap states and safety relay states.

0x01 Front flap sensor.

0x20 Left flap sensor.

0x40 Right flap sensor.

0x80 set to 0

0x10 set to 0

0x20 set to 0

0x40 Safety relay #1.

0x80 Safety relay #2.

Unsafe condition is 0, safe condition is 1 (hence 0b11000111 = all OK)

TEMP

Returns the temperature reading of the temperature sensor mounted on the Spot PCB.

BUTTON 1

Front panel On-Off button will be red when powered up (default state)

BUTTON 0

Front panel On-Off button will be blue when powered up. Button colour option is saved in EEPROM.

BI#

Starts the firmware boot load process. Details as for VSC6000/8000.

You may need to add a 2-3 second delay between opening the FTDI driver and sending this command.

Note: If boot load fails or is interrupted, link pins 3 and 5 on the PROG connector and use the Windows BootLoader application (V4 mode). Remove the link after the firmware update.

WARNING: Do not press the front panel on-off button or remove power during a bootload process!

Z#~

Resets the main V80 microcontroller. You are advised to close all USB devices connected to the hub, since the power will be removed from the hub within 250 milliseconds. You can increase this time delay with **Ky=n** which will set the default time delay to n milliseconds.

Note there is no newline character returned after this command.

BEEP a, b

Sounds the main controller piezo buzzer. The pitch half period is **a** microseconds, the duration is **b** milliseconds. **BEEP 1073, 200** will sound B flat above middle C for 200 milliseconds.

BEEPX

A variation on the above. No parameters required. (from BWV 785)

SE=1

If the FCB camera returns an error packet, a short beep will be heard. Use this command to suppress the beeps. **SE=0** will enable beeps, this is the default power on setting. See the FCB manual for an explanation of VISCA error packets.

Kt=n

Sets the delay time between issuing a processor reset instruction and the reset. This delay time is saved in EEPROM. Also applies to the Kill command.

Kt?

Reads the reset delay time.

KK

Sends a 'kill' command to the main VSC80 microcontroller after the delay time set by **Kt**. This has the same effect as holding down the front panel button for 2-3 seconds. Note there is no newline character returned after this command.

Power supplies, etc. for testing.

Defaults to on at reset/power up.

K3=1 3V power on

K3=0 3V power off

K5=1 5V power on

K5=0 5V power off

Kc=1 Camera power on

Kc=0 Camera power off

Ut=1 FTDI interface on

Ut=0 FTDI interface off

U2=1 USB 2 hub enabled

U2=0 USB 2 hub disabled

U3=1 USB 3 hub enabled

U3=0 USB 3 hub disabled

ENABLESHUTDOWN=n (VSC80i only)

If n =1 will send a shutdown signal to the internal PC when the front panel button is pressed. This setting is not remembered and will need to be sent at power on if required. When the VSC80i firmware has detected the power down state of the internal PC it will also switch off.

The front panel button must be released immediately else it will remove the power to the internal PC within a few seconds.

Left and Right Flood PCB

Two boards, each consisting of three UV, two OVD and two White LEDs. The White LEDs also double up as additional OVD sources (in OVD 1-4 and 18-21 group). A maximum of two of these LEDs on each board can be illuminated at any time. The two OVD LEDs are controlled with the OVD command.

Additionally there are four pairs of IR LEDs, two for each wavelength. Any number of these may be on. The following commands operate on both boards together. To turn off or dim either side, use a dimming control.

PWMP n

Sets the PWM controller prescaler byte. Default 7 (830Hz), minimum 3 (1660Hz).

UV2 n

Turns on the three additional UV LEDs at intensity n.
($\lambda = 385\text{nm}$)

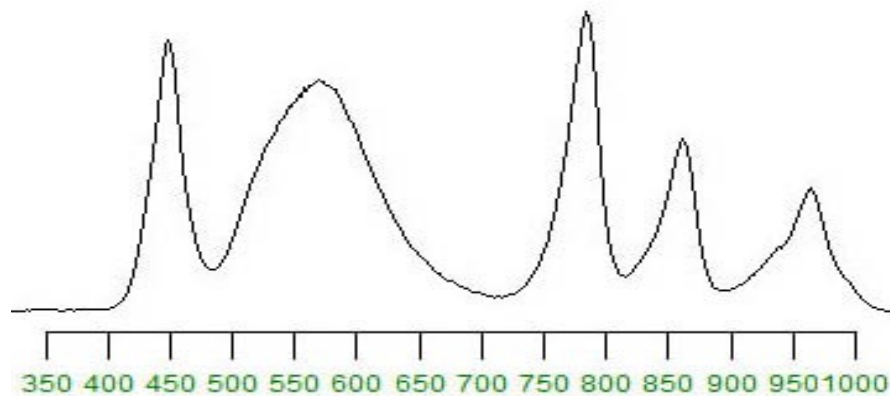
IR780, IR850, IR940, IR980, W1 and **W2** turns two LEDs (for each colour) at intensity n.

W12 n

Turns on both (W1, W2) white LEDs together. Note that W12 does not turn off the previous lighting. It may therefore be used to provide dimmed background illumination while other sources are on. It will also provide flicker free dimming if n is varied. The DAC setting for this defaults to 1000. Use $P1=n$ to reduce this. At very low values of both P1 and W12, the LEDs may flicker.

IR n

Turns on IR780, IR850, IR940, IR980, W1 and W2 at intensity n. Typical emission spectrum, read with VSC80 spectrometer, without intensity calibration:



IRX 1100 n

Turns on up to four IR flood LEDs together. In this example, 780nm and 850nm on at intensity n. There must be exactly four 1s or 0s in the string to represent the four IR LEDs corresponding to the order 780nm, 850nm, 940nm and 980nm. If n is not specified, the intensity is not changed from the previous value. All LEDs use a common intensity value. For all the above, $n = 0$ will turn off whatever is on.

IRW 110011 n

Turns on any number of IR and white floods. In this case 780nm, 850nm, White1 and White 2. See above for more details.

Intensity level control is by pwm at 790Hz. The hardware supports individual dimming of each LED channel, but this is not (but could be) supported in firmware. The intensity parameter is currently applied to all LEDs.

Left and Right Flood PCB

Multi-Spectral Version (for VSC800)

This board replaces the left and right flood boards and adds additional flood leds of 16 wavelengths 405 nm to 890 nm.

MS n

Turns on multi spectral LED n

n is ordered as 17 wavelengths:

365, 405, 425, 450, 470, 490, 520, 560, 600, 620, 640, 660, 700, 720, 750, 810, 890.

365 nm is identical to the original UV LED on the flood board.

If this board is fitted in place of the original flood board, the ENUM command will return "LFLOOD", "RFLOOD" and additionally "LMS" and "RMS" for left and right respectively. The original flood board only returns "LFLOOD" and "RFLOOD"

This board also has the original White, UV, OVD (1-4, 14-17) and IR LEDs listed on page 6. The commands for these are unchanged.

The firmware which includes the multi spectral addition should also run if the original flood boards are fitted, but the MS command will not be available.

Spot PCB

Nine coloured LEDs and one White LED.

VIOLET n

Turns on violet spot LED only at intensity n. (0-1023)

BLUE, BLUEGREEN, CYAN, GREEN, AMBER, RED, DEEPRED, FARRED, WHITESPOT.

As above to turn on one only of these LEDs.

SPOT 1100000000 n

Turns on Violet and Blue spots LEDs together at intensity n. There must be exactly ten 1s and 0s in the first argument string to turn on any number of LEDs corresponding to the LED order from left to right of: Violet, Blue, Blue-Green, Cyan, Green, Amber, Red, Deep Red, Far Red and White.

If n=0, LEDs will turn off.

If more than 2 LEDs are on, the intensity level will be automatically de-rated, depending on the number of active LEDs. The maximum de-rating factor for all 10 LEDs is 0.5 . The actual intensity applied will be returned. If **P4** is subsequently used to adjust the intensity, the applied intensity, de-rated if necessary, will also be returned.

The de-rating factors are, for 1 to 10 LEDs:

1.0, 1.0, 0.90, 0.80, 0.75, 0.70, 0.65, 0.60, 0.55, 0.50

LED wavelengths are:

Violet	410nm
Blue	445nm
Blue/Green	475nm
Cyan	500nm
Green	520 nm
Amber	590nm
Red	640nm
Deep Red	660nm
Far Red	740nm
White	420nm-700nm

Lamp Tray PCB

Sidelights, OVD, Coax (Lamp tray PCB)

21 OVD LEDs, 1 Coax LED, Left and Right White and IR LEDs, Two UV tube inverter drivers (312nm and 254nm)

Lq = 1, Lq = 2, Lq = 3, Lq = 4

Turns on one of OVD1 - OVD4 LEDs located on the Left Flood PCB

All OVDs run on the same scale of 0 (off) to 1023 (maximum)

Lq = 5...13, Lq = 18...22

Turns on one of 13 OVD LEDs on the Lamp tray PCB

Lq = 14, Lq = 15, Lq = 16, Lq = 17

Turns on one of the OVD14 - OVD17 LEDs located on the Right Flood PCB

To specify the OVD intensity, add a second parameter:

Lq=5 200 will turn on OVD5 at intensity 200. The intensity level for each OVD led will be saved in EEPROM with this command. If a subsequent OVD command has no level parameter specified, then the value stored in EEPROM for each LED will be used.

Lq? n

Diagnostic to read the EEPROM level stored for OVD LED n

Arrangement of OVD LEDS is:

					22							
4	5	6	7	8	9 (21)	10	11	12	13	14		
3					20					15		
2					19					16		
1					18					17		

Lamp tray continued...

COAX n (Ax=n)

Turns on Coax LED at intensity n

SIDELEFT n (Ls=n)

Turns on Left White and IR side LEDs together at intensity n.

SIDERIGHT n (Rs=n)

Turns on Right White and IR side LEDs together at intensity n.

SIDE 1100 n

Turns on one or more side lights at intensity n.

The first parameter is a string of exactly four 1s or 0s which controls the side lights in the order:

Left White - Right White - Left IR - Right IR

The example above will turn on both the left and right white LEDs and turn off both IR LEDs. This may not be a lighting requirement, but it allows individual LEDs to be tested.

UV254 n (Uf=n)

Short wavelength UV tube.

UV312 n (UV=n)

Medium wavelength UV tube.

UV tubes are either on (n != 0) or off (n = 0)

See also the **Uv2** command on the Flood boards for UV LEDs.

Transmitted light PCB

TRANSSPOT n (Sx=n)

Turns on transmitted IR and White Spot LEDs at intensity n

TRANSWHITE n (Ix=n)

Turns on transmitted IR and White flood LEDs (850nm) at intensity n

TRANSUV n (Ux=n)

Turns on transmitted UV floor LED (385nm) at intensity n

T1=n, T2=n, T3=n, T4=n can be used to turn on the above white and IR LEDs individually. This may not be a lighting requirement, but it allows individual LED banks to be tested.

T1: White spot

T2: IR spot

T3: IR flood

T4: White flood

Lighting levels.

Various PCBs have their own local intensity control for the LEDs. If the intensity parameter is not given in the above commands, then the previous value will be used. Most are DACs, but the tray LED level is set by PWM. (there may be differences in level numbers if OVDs are dimmed since some are DAC and some are PWM).

Should it be required to dim individual LEDs, these are available:

P1=n

Sets the Left flood board intensity. Applied to all LEDs on this board.
See P6, P7 for separate visible and UV levels.

P2=n

Sets the Right flood board intensity. Applied to all LEDs on this board.
See P8, P9 for separate visible and UV levels.

P3=n

Sets the Transmitted board intensity.

P4=n

Sets the Spot board intensity.

P5=n

Sets the Lamp Tray board board intensity. (L & R side lights, Coax, OVD 5 – OVD17).

P6=n

Sets Left flood board visible intensity.

P7=n

Sets Left flood board UV intensity.

P8=n, P9=n

As above for the Right flood board visible and UV intensity.

Note that OVD1 – 4 are on the Left flood board, OVD 18 – 21 are on the Right flood board and OVD 5 – 17 on the Lamptrey board. The **Lq** command will allow individual levels to be specified for each OVD and should be calibrated with this command – the OVD array has two different dimming schemes, so it is preferred that the saved intensity is always used, rather than the above commands.

Motors

C1

Camera filter wheel. (There is no C2 camera filter motor fitted)

C1=n, C1?, C1>, C1<, C1h, C1z, C1o

Operates as V40 filter wheel.

C1=n moves the filter wheel to position n (0-12), by the shortest route.

C1? Returns the filter position.

To calibrate: Apply C1z to clear the calibration. Apply C1h to home the wheel.

Use C1> or C1< as required, to nudge the wheel either direction to centralise the clear filter element in the field of view. This adjustment value is stored in EEPROM. Finally apply C1h to home the wheel, and verify that the alignment is OK.

C1h can be applied thereafter anytime. Motors must be homed from a home command at power on.

C1o is a diagnostic to read the offset value and will return a signed number as a string.

C1F

Returns filter population string. 1 = filter element is fitted, 0 = filter element blank. (future use)

C3

Close up lens.

C3=n, C3>, C3<, C3h, C3z

To calibrate:

Apply C3h moves motor to find sensor (out)

Apply C3z stores 80 degrees in memory. (about 350 steps)

C3=1 Moves to above value.

C3>, C3< Nudges above value and motor, then saves value in memory.

Thereafter use C3h (or C3=0) to move out and C3=1 to move into the optical path.
The close up lens will not move until it is calibrated.

C4

Coax mirror Motor.

Required to be in position when Coax source is on, or parked otherwise.

There are two calibration points, one to set the filter horizontal (parked) and the second to set the filter to 45 degrees, or as required for a coax image.

Calibrate as for camera filter. Apply: C4z, C4h, then C4> or C4< as required to set horizontal. Apply C4=1, then C4p or C4m as required for a coax image (a suitable coax sample is required). Finally apply C4h (or C4=0) and C4=1 to verify both positions are correct .

The coax lens will not move until it is calibrated.

C4z Clears both the calibration points.

C4h Home (parks) motor(horizontal).

C4>, C4< Fine adjustment to home position.

C4=1 Move to working position (45 degrees or as required)

C4p, C4m Fine adjustment to working position.

C4=0 Equivalent to home (park).

Cnh will home all three motors together.

Spare motor (not connected)

C5=n, C5?, C5>, C5<, C5h, C5z

Anti Stokes flash PCB

Photo flash tube and charging circuit.

CHARGE=**n** (**FI+**)

If **n** is 1, turns on the charging circuit (allow approximately 5 seconds to fully charge).

CHARGE=0 or **FI-** Turn off the charging circuit. Firing the flash will also turn off the charging circuit. This function can also be used to remove the freeze frame state, the camera then outputs live video. Note the charging circuit should also turn off when a fully charged state has been detected by the charging electronics.

FSTATUS (**FI?**)

Returns 0 when fully charged, else returns 1.

FLASH (**FI@**)

Fires the flash, then freezes the frame after the number of frame counts set by **Fd=n**. Use **FI-** to turn freeze frame off. **Fd** is set to default to 1, which probably works. If not try different values. The flash function should return zero. If non-zero is returned it indicates that the VSC80 could not detect the camera's horizontal sync pulse and a timeout condition was detected.

LL@ (Phosphorescence?)

Waits for the next camera frame sync pulse then delays by the number of milliseconds specified by **Gd=n**, then switches off all LEDs with a Global Enable signal. This function will capture and freeze the video after a further frame count set by **Ld=n**. Where **n** can be 0, 1, 2... selected by experiment. This function should return zero. If non-zero is returned it indicates that the VSC80 could not detect the camera's horizontal sync pulse and a timeout condition was detected.

After use, you may need to re-enabled Global enable with **Ge=1** to turn LEDs back on, unless there is a LED change or LED on command, which should automatically re-enable the Global enable. You may also need to clear the controller buffer which holds the VISCA reply packet using the command **FCB**, which returns and clears the return buffer in the microcontroller. For normal VISCA commands, the buffer content is returned and cleared automatically.

Gd? and **Ld?** Will return the above settings. One frame is 40 milliseconds. **Gn** should not normally exceed 40. The camera frame rate may need to be 1/25 (40 milliseconds) for Anti-Stokes and phosphorescence to operate correctly.

EEPROM

The VSC60 has a user available 32k EEPROM organised as pages of 64 bytes.

Xb=n

Selects the page number (0-510)

Xr?

Read 64 bytes from the current page. After sending this command, read back 64 more bytes. A newline character is not returned after the 64 bytes are received, so make sure exactly 64 bytes are requested.

Xw=string

Writes a string up to 64 characters in length to the current page. If the length is less than 64, the page will be zero padded. String may contain CR characters, (0x0d) but not LF (0x0a) which terminates the command. A read verify is done after the EEPROM write, this command will then return the number of bad bytes, which hopefully will be zero.

Xw+

Writes 64 bytes to the current page:

1. Send **Xw+**
2. Wait for a newline character to be returned.
3. Write exactly 64 bytes.
4. Read back two bytes. The first byte should be null (0x00) followed by a newline. If the first byte is not null, an error occurred. The write data is checked against a read of the data, the first byte returned is the number of errors detected.

Note that write data should be limited to 7 bit ASCII.

Machine settings (motor calibration, firmware options etc.) are stored in a different area of EEPROM inside the microcontroller. Firmware upgrades do not overwrite any EEPROM data.

VISCA Command and Inquiry (VSC80 only)

You can send and receive VISCA packets to the camera by sending the VISCA command packet directly to the V80 controller. The command and inquiry packets are as given in the camera Technical Manual, followed by a newline character.

For example "**8101040102FF\n**" will turn on the ICR (near infra red) mode. The returned packet should be **9041FF9051FF** (acknowledge packet followed by completion packet)

Inquiry packets are handled in the same manner, "**8101044AFF\n**" will return the shutter value. For example "**905000000005FF**" is returned when the shutter is at index position 5 (1/25).

To prevent lock up in case a return packet is not received from the camera, zero will be returned after a timeout period. This timeout can be set by **TO=n**, the units of n are 65 milliseconds. The default setting is 8 (8*65 ~= 0.5 seconds).

FCB

Will return the last reply from the camera, which is retained by the V80 controller. If this buffer has already been read (it usually has in most cases), zero will be returned.

Reading this value at 1 second intervals at start up may indicate when the TWIGA board has completed sending VISCA initialisation commands. Wait for 2 consecutive zero responses.

Additional VSC800 Commands

FTDI - USB interface to PC. Open the FTDI driver by Serial Number "VSC800".

Zoom Lens motors

Cc=n

Moves lens focus to position n. The firmware will limit the values of n to those given by Cch and Ccl

Cch

Returns the maximum lens focus position.

Ccl

Returns the minimum lens focus position.

The position range of all lens motors is less than 0 to 1023. (The prototype gave c: 25-1002, z: 23-1002, i: 68-973)

Cc?

Returns the current position of the focus motor. It may still be moving!

Ccs

Returns the status of the focus motor. It should be zero if there are no errors. Errors will usually be due to a timeout, the motors will turn off if failed to reach their position after 130% of the expected full travel time.

Cct

Returns the travel time, in milliseconds, for the last focus motor move. Full range travel times are approximately 3000, 2700 and 1700 for focus, zoom and iris respectively.

Cmm

Returns zero when all lens motors have completed any movement.

Ccc=n

Moves the focus to position n. If n is less than the current position, the motor will move back, then forwards to the requested position to remove backlash. This should give a more repeatable position.

Note move commands will return immediately, the motors will be driven to their destination position under interrupt control. All 3 may be moved at the same time if required.

The above commands can be applied to the Zoom and Iris motors by replacing all the lower case characters 'c' with 'z' and 'i' respectively.

Lh

Can be used to home all three lens motors at the same time. Motors will also home on power up.

Camera filter

C1=n, etc.

Camera filter operations (as VSC80). There are 15 filter positions.

C1B will report the number of filter positions in the filter wheel.

C3..... There is no close up lens assembly fitted to the V800. The camera filter wheel is populated with a close up lens in the last position.

Back focus

Bc=n

Move back focus to position n.

Bc?

Report position. Maximum position is 24,000.

Bc+,

Move back focus motor Forward.

Bc-,

Move back focus motor Reverse.

Bc#

Stop back focus motor. Motor will stop at limits (0-24,000).

Bcz,

Clear the home offset.

Bch,

Move to home+offset.

Bc>

Adjust home and offset position forward.

Bc<

Adjust home and offset position reverse.

SSTATUS (S??) (motor fault bits set if failed to find the home sensor)

Returns the status of the motors.

0x01 Camera filter motors not yet home.

0x02 V80 Close up lens motor not yet home. (**Back Focus in V800**)

0x04 One or more motors are moving.

0x08 V80 Coax motor not yet home. (**Not used in V800**)

0x10 Camera filter motor fault.

0x20 V80 Close up lens motor fault. (**Back Focus in V800**)

0x80 V80 Coax motor fault. (**Not used in V800**)

Reduced Specification Machines

Machine type **M1** and **M2** have a reduced number of components. The following commands apply.

Op?

Returns the machine type.

0: Full specification VSC80 (FS).

1: M1

2: M2 (4 spot LEDs, 6 Camera filters, no Anti-Stokes)

C1p

Returns a 13 character string. '1' indicates the camera filter in that position is present, '0' indicates there is a blanking plate in these positions.

FS machines will return "1111111111111".

M2 machines will return "1001001010110". The left most character is filter position 0 (visible filter), the right most is filter position 12. **C1=n** will not move the filter wheel to blank positions.

It may be assumed that machines other than FF, M1, M2 may exist in the future, due to the unpredictable nature of Sales.

Note: Machines built before the M2 change will return the filter string with **C1f**. If it returns all 1's assume it is a FS machine.

Shp

Returns a 10 character string. '1' indicates that the spot wavelength LED is fitted and '0' the LED is not fitted.

FS machines will return "1111111111".

M2 machines will return "0100101001". The wavelength order is left to right violet through to white, so Blue, Green, Red and White Spot LEDs are fitted.

As?

Returns 1 if the Anti-Stokes unit is fitted.

M2 machines will return 0, all flash gun commands will not be available.

The above information is stored in EEPROM memory.

The following may be used to configure a machine. Production will be supplied with a Windows utility program to program the settings below:

SETMACHINE n

For FS machines n should be set to 255. For M1 machines, set n to 1, for M2 machines set n to 2.

SETFILTER 111111111111

Will set the filter wheel to type FS.

SETSPOT 1111111111

Will set the spot LEDS to type FS

SETSTOKES n

For machines with Anti-Stokes fitted, n should be 255, for machines without Anti-Stokes, n should be set to 0.

I2C Address

Various components are connected to the main processor with an I2C bus. You can check the acknowledge from these devices by sending the command **I2C n** where n is the 7 bit device address shifted left by one bit (the write bit is 0). Values for n are (use the hex or decimal form):

Temperature sensor (on Spot PCB)	0x8e	142
Transmitted LED base	0x46	70
Transmitted LED base DAC	0x24	36
Left flood	0x86	134
Left flood DAC	0x20	32
Right flood	0x98	152
Right flood DAC	0x3e	62
Spot PCB	0x5a	90
Spot PCB DAC	0x3c	60
Flash unit	0x40	64
Lamp Tray	0x50	80
Error	0x0d	13

All these should return zero when they respond, except Error which is an address which does not exist. If this returns zero, the data line may be stuck. A missing or faulty device usually responds with -2 (try a re-boot of the system!)

Diagnostics, etc.

DAC0 n

DAC1 n

Sets Flood DACs to n

I2C byte sequences can be sent directly from the command box (if you can work them out!). Mainly for firmware/hardware bash testing:

OUT 72 20 -73 r

This sends the I2C byte sequence:

72 (0x48) The flood I/O expander Write address.

20 (0x14) The address of the Latch A register.

73 (0x49) The I/O flood expander Read address, preceded by a restart state (-).

r Byte read request. Returned byte is in the **Reply** box
(reads the contents of register 0x14 on the flood PCB expander)

OUT 72 20 n

Writes n to the I/O expander Latch A register.

I2C address

Writes address byte to I2C device. Returns zero if device acknowledge, else returns non-zero. Address may be decimal or hex if preceded with 0x

Lamp Tray I2C Commands, using OUT (6 bytes)

80 0 P H L 0

80 = Slave address, P = OVD number (1-13), H = upper 8 bits of pwm , L = lower 2 bits of pwm

80 0 14 0 0 0

OVD Off.

80 0 15 H L 0

Coax LED On.

80 0 16 0 0 0

Coax LED Off.

80 0 17 P H L 0

Side Lights pattern P (4 bit pattern) on.

80 0 18 H L 0

Sets pwm level.

80 0 19 H 0 0

If H, turns on 254nm tubes.

80 0 20 H 0 0

If H, turns on 312nm tubes.

80 0 21 0 0 0

Coax mirror in working position.

80 0 22 0 0 0

Coax mirror parked.

80 0 23 0 0 0

Nudge park position forward.

80 0 24 0 0 0

Nudge park position reverse.

80 0 25 0 0 0

Home (park) coax mirror.

80 0 26 0 0 0

Erase coax motor calibration. (sets working position to 45 degrees)

80 0 27 0 00

Nudge working position forward.

80 0 28 0 00

Nudge working position reverse.